

Understanding Expert Crafting Practices of Blind and Low Vision Creatives

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ABSTRACT

Recent research has investigated how to make digital tools accessible to blind creators, focusing on visual manipulation using gestures and other input modalities. But while prior work centers on the inaccessibility of content creation tools, less is known about how blind and low vision creators engage in analog creative activities. We explored how expert blind and low vision creatives used a variety of craft-specific and common household materials in creative activities. We interviewed vision impaired crafters about materials and tools they use, their creative practices, and their collaborations. We found that vision impaired crafters employed strategies to accessibly organize materials and that they solicited feedback from mixed-ability collaborators. These insights contribute knowledge that can inform how to structure mixed-ability design collaborations and technologies to be useful for blind and low vision participants.

CCS CONCEPTS

• **Human-centered computing** → Accessibility; Accessibility design and evaluation methods; Human computer interaction (HCI); Empirical studies in HCI.

KEYWORDS

Blind and Low Vision Creatives, Crafting, Design

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1 INTRODUCTION

Early work investigating content creation systems for blind drawers identified the drawing capacity of blind and low vision people as comparable to sighted people's drawing skills, specifically when it comes to generating imagery to represent familiar environments [8]. In particular, mental models of sighted and blind drawers were much the same when it came to replicating objects and positioning in space, as long as all parties had access to referents of items in

the environment [8]. Despite this understanding of how blind and low vision people can draw, it is unknown how to leverage their current creative skills, including and beyond drawing, to inform the design of content creation tools [12, 14].

Harnessing the drawing skills of blind and low vision creatives could inform inclusive creation practices and applications, but most work to date has focused on auditing and developing tools for digital content creation [7, 10, 15–17], and has not examined analog efforts of crafting experts. We explored the creative process in non-digital crafting activities of blind and low vision individuals to better understand how they engage creativity and collaborative feedback. In particular, our study was focused on: (1) What materials and tools do blind and low vision creatives use in their crafts? And (2) What techniques do they use in their creative endeavors and in collaboration with sighted colleagues? To address our research questions, we interviewed blind and low vision adults about their experience in crafting.

2 RELATED WORK

2.1 Support for Visual Drawing for Blind and Low Vision People

Blind and low vision people have some experience creating digital content, but are stymied by the inaccessibility of tools [10, 12, 14]. To address these limitations, researchers have explored tactile and audio/acoustic perception of visual imagery to enable blind people to access drawings and figures [8]. Specifically, researchers considered that certain aspects of imagery are important to blind people: such as sides and outlines, but probably not colors and shading. In particular, “since the referent of both the mental model of the blind and the mental model of the sighted person is the same (the real world), we can conclude that they have about the same model as sighted people (except for some obvious features such as colour)” [8]. In other words, given the same referent, blind and sighted people can perform the same tasks to represent objects in a given environment.

Research investigating blind drawing in the context of drawing support focused on drawing without visual feedback [7]. But, to understand how blind people perceive and make sense of visual rendering, such as images or user interfaces, it helps to understand how they create representations of their mental model using their own blindness skills or other approaches. Indeed, less is known about how to support blind and low vision directed content creation [7, 15].

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2.2 Creative Ideation using Participatory Design with Blind and Low Vision Designers

To capitalize on blind and low vision people's creative abilities, previous work used participatory design to explore accessible ideation, in which mixed-ability groups of participants used different supplies and strategies to brainstorm ideas [2, 4]. Researchers identified that low-cost household items coupled with accessible group practices (e.g., only one person speaks at a time) productively facilitated shared group creative brainstorming [2]. In contrast, group dynamics created challenges for non-dominant individuals to contribute equitably [4], and varying mixed abilities made it difficult to share ideas using only tactile artifacts [2]. Expanding on tactile ideation, researchers explored scenario-based and prototyping approaches to facilitate idea generation, but such work used interventions (scenario-based prompts and low-fidelity prototyping technique [1, 11, 13]) to elicit ideas among blind and low vision participants. Findings from these endeavors showed effective ways of engaging blind and low vision people in design, but the activities used originated from design technique, rather than learning from what blind and low vision creatives already do.

Aside from exploring creative practices, prior work showed that low-cost and easily accessed materials were effective in ideating and prompting discussion [2, 4]. These materials included clay, pipe cleaners, paper clips, and LEGO figures. Physical and tangible forms can explore ideas as well as 2D drawing approaches, and commonly used craft tools appear to be sufficient to ideate different design ideas [2, 4]. However, this prior work seeded low-cost materials to determine their effectiveness, meanwhile we seek to understand what materials blind and low vision creatives *already use* in their creative pursuits. In particular, although others have focused on creative aspects like art appreciation [9], and producing accessible presentations [12, 14–17], we adopt the HCI approach of learning from analog practice to understand existing skill and approaches to creative endeavors [6]. Specifically, Hinckley et al. observed users' manual pen and paper manipulation to inform the design for digital stylus and tablet interactions [6]. Similarly, with this work we attempt an understanding of existing strategies and practices of blind and low vision people to inform the space of creation.

3 METHOD

We first distributed a questionnaire to inform our interviews, asking blind and low vision individuals about their creative pursuits and non-audio-based materials they used in creative activities. We asked about non-audio-based materials to learn about analog and tactile methods, rather than digital ones. We drew on questionnaire responses to outline interview questions. Then, we conducted remote interviews with 18 vision impaired crafters about their creative processes. This work was approved by the Rochester Institute of Technology's Institutional Review Board.

3.1 Phase 1: Preliminary Questionnaire to Understand Material Use

Before diving into interviews, we sought to inform ourselves about the range of materials and crafts blind and low vision people engage. We created a Google Forms questionnaire consisting of six questions and distributed it to blind and low vision participants via a local

blind and low vision organization and by sharing through our professional networks (see Appendix I). We asked about crafting endeavors and materials used, including (1) craft specific materials (open ended question) and (2) common household craft materials (select from a list). Since we could not know about crafts beforehand, (1) enabled them to tell us what they used unique to their craft, e.g., knitting needles. Based on Bennett et al.'s work [2], we were curious to know if common household might also be used. To aid recall for such items, we listed items for selection with an option to add other: cardboard, felt, construction paper, index cards, foam boards, popsicle sticks, cork stoppers, clay, rubber bands, cotton balls, LEGO blocks, Play-Doh, pipe cleaners, scotch tape, and nail polish. The questionnaire took about 3 to 4 minutes to complete.

We collected responses from seven individuals (Table 1). This small sample size was sufficient to provide enough information to ensure our questions were informed and appropriate. We refrain from making claims about how many people use specific materials based on questionnaire results. All respondents self-identified as legally blind or low vision due to severe vision impairments. Five participants had previous experience in crafting activities (Table 1). Respondents indicated a variety of materials used, including thread for embroidery, beads and charms for jewelry making, and wood and table saw for woodworking. Respondents used all common craft materials in our list, except toothpicks.

3.2 Phase 2: Interviews

In phase 2 of the study, we conducted semi-structured remote interviews with vision impaired crafters who were over 18 years of age and had experience crafting as a profession or hobby. Participants were recruited through the local blind and low vision community, professional networks, and snowball sampling.

Interviews focused on practices in crafting activities similar to sketching, such as ideation, eliciting feedback, and iteration (see Appendix II). We recruited interview participants with above novice level of crafting experience, determining level of expertise based on self-reported competence and length and intensity of experiences. Preference was given to participants who (1) had been practicing a craft for a minimum of two years as they were more likely to have experienced creative ideation and iteration by eliciting feedback from others; and/or (2) gained skills in a new craft through intense and frequent practice, such as during isolation due to COVID-19. We later broadened the range of experiences due to recruiting challenges, and included participants with crafting experiences from two and a half months to 15 years.

Eighteen participants (12 women) from the United States (15), Canada (2), and Portugal (1) were interviewed. Ten identified as legally blind and 8 identified as low vision. Despite our best efforts, we were unable to recruit participants who identified as blind. Participants had experience in a wide variety of crafts, including knitting, crochet, metal fabrication, LEGO building, mixed media art, eyeglass frame customization, costume making, ceramic, and so on (see participant details in Table 2). Interviews were conducted via video or audio conferencing software, were 60 to 90 minutes (mean=83 minutes), and were recorded and transcribed. Each participant received an electronic payment or money order of \$30.

Table 1: Details of Questionnaire Respondents

Participant	Self-reported Visual Ability	Crafting Activities
P1	Zero vision (no light/shadow perception)	Sound effects mixing and collecting, and playing hand drums non professionally.
P2	Totally blind	<i>"At this time none, but I would like to."</i>
P3	Legally blind	None.
P4	Legally blind (<i>"I can see light and shadows of people but I can't really see their facial expressions, eye color, skin color or hand gestures."</i>)	Making slime with different ingredients and colors, jewelry making, and weaving.
P5	<i>"I have had retinitis pigmentosa since I was 18. I see light, shadows, and movement. very light sensitive."</i>	Woodworking.
P6	Very severe cataracts in both eyes	Embroidery, collage, paper sculpture, jewelry making, sculptural basketry.
P7	Blind in one eye, legally blind in the other	Painting, drawing, digital art, costuming.

Prior to each interview, we sent consent information and asked participants to send photos of their creations so that we could ask informed questions about their craft. Based on results from the questionnaire, interview questions centered around crafting activities, materials used, creative process (especially changes and improvements made based on feedback received), and experiences with collaboration, if applicable. We asked about accessibility challenges. We approached blind and low vision participants as experts in creative activities to garner insights into how they used materials and engaged in creative activities. We followed up about common materials that were not craft specific, sometimes by referring to materials reported in the preliminary questionnaire. Some participants demonstrated their craft, and showed crafting areas, materials and artifacts. We took screenshots of these views with their consent.

We transcribed interviews using Otter.ai, and edited for clarity and accuracy by listening to audio recordings, reviewing video, and referring to photos of artifacts sent by participants. One researcher conducted a thematic analysis on the data using open coding, axial coding, and selective coding approaches [3]. During open coding, we first created tentative labels to describe individual chunks of data to highlight instances relevant to crafting and the creative process. These open coding labels were further refined based on properties and meanings of succeeding interview data. As we identified connections and relationships among the open codes during axial coding, relevant codes were grouped together and further categorized into prominent themes. Interview data and analytic themes were reviewed again with the research team to ensure relevance and accuracy to the data.

4 FINDINGS

We identified four themes from the interview data: (1) selection and organization of tools and materials of blind and low vision individuals in crafting activities; (2) creative iteration in different forms; (3) feedback solicitation and iterative improvements; and (4) how blind and low vision individuals engage in creative collaboration.

4.1 Selection and Organization of Tools and Materials

Participants used a large variety of tools and materials across a wide range of creative pursuits (see Appendix III for the full list). They primarily used craft specific tools like different types of stabilizers for machine embroidery, different sizes of crochet hooks, pottery wheels for ceramic making, and computer numerical control plasma table for metal cutting and fabrication. They used everyday household objects, such as toothpicks, chopsticks, nail polish, and cooking tools.

Participants used assistive devices to access these tools and materials, such as magnifying tools and large print. For example, P1 read patterns and marked fabrics with magnifiers; P17 used large print crochet tutorials; P5 used digital LEGO instructions for zoom capability. Yet, digital tools could be costly and not sufficiently accessible, as P17 mentioned, such tools could add challenges, *"my screen reader just goes too fast. . . Because I have it set at like 200 words per minute, and that would just be impossible to try to keep up stitch by stitch."* Some participants used digital tools to aid workflow efficiency. P13 used a computer-aided design program in metal fabrication to straighten and center because he was not confident about his residual vision.

Participants used specific criteria when choosing materials, including: objectives of the craft (seeking materials that expressed their creativity with look and feel, e.g., P12 used clay instead of real shells because *"I didn't want them to crack, like if it falls, cause real seashells are fragile. So I thought that the clay would be more shatterproof"*); availability (using what was at-hand, e.g., P11 mentioned, *"I buy yarn that I like and then I pick a pattern and see if the yarn will work"*); tactility (texture and feel enhanced orchestration, e.g., P5 liked LEGO, *"it's so inherently tactile, it sometimes just feels really nice to just start putting things together and see what you get"*); and tentative and temporary characteristics that allowed them to experiment with options (P12 explained that nail supplies were less permanent than acrylics which enabled her to make changes, *"if I don't like the color, I can just grab some nail polish remover, and take it off"*).

Table 2: Details of Interview Participants

Participant	Gender	Self-reported Visual Ability	Crafting Activities	Experience Level of Primary Crafting Activities
P1	F	Legally blind	Sewing	"I think since like, probably 2005? maybe 15 years?"
P2	F	Legally blind	Multi-medium art, charcoal and pastel drawing, acrylic painting	Intermediate in multi-medium art.
P3	F	Legally blind	Decoupage, multi-medium crafts, dream catcher making, drawing, pottery	(Multi-medium crafts) about 5 years.
P4	M	Legally blind	3D multi-medium crafts, pottery	3 months but had experience as a child.
P5	M	Low vision	LEGO creations	"I... have built LEGO sets and creations all my life."
P6	F	Low vision	Embroidery	Between novice and intermediate (A year and a half; <i>"Since then I've had over 300 Etsy sales."</i>)
P7	M	Totally blind	Tactile drawing and Braille dot drawing	Novice ("I think I first started learning about tactile pictures and seeing them as a really young kid.")
P8	F	Low vision	Furby toy customization	Advanced (Roughly 3 years).
P9	F	Legally blind	Modular origami, crochet, knitting, pottery	Intermediate in modular origami.
P10	M	Low vision	Tech crafts, video production	Advanced.
P11	F	Low vision	Knitting, crochet, spin, weave, and dye	Advanced in crochet and knitting.
P12	F	Legally blind	Ornate box creation, eyeglass frame customization, Barbie customization, jewelry making	<i>"(Making ornate boxes) probably since about high school, which I graduated in 2009."</i>
P13	M	Legally blind	Metal fabrication	Advanced.
P14	F	Legally blind	Cold-press soap making	Intermediate.
P15	F	Legally blind	Knitting, cosplay costume making, painting, sewing, digital art	Intermediate in knitting.
P16	M	Legally blind	Pen and ink drawings, digital art, spray painting	Advanced in drawing.
P17	F	Low vision; blind in left eye	Crochet	Between intermediate and advanced.
P18	F	Low vision	Crochet, embroidery, sewing, ceramics	Advanced in crochet.

Participants developed organization systems to manage and locate materials. They used storage containers like totes, plastic containers, trays, bookcases, and utility carts to store materials. Inside these containers, they sorted materials for quick selection. P15 kept knitting needles in a small bag within a bigger container with strings and yarns for knitting projects. Participants developed workarounds to identify materials in storage containers. P8 labeled the outside of her boxes with "tactile dots," raised dots made of hot glue, because Braille dots were too expensive. She assigned meaning to these tactile dots: *"with two dots, this is my paint box. Or this one with three dots, this is my cosmetic box."* P13 marked his wrenches with big blobs of texture paint so that he could feel them right away when reaching into toolboxes.

As they worked on projects, participants put out materials needed for the specific project for quick access. P1 reported, "I keep like all my stuff in a little tray, my scissors, my pins, whatever

I'm working on for just that one project, and then instead of having to scan the entire dining room table and spend half an hour trying to find one thing." P5 emphasized the importance of space between piles of LEGOs although he did not specify how far the piles from each other would be ideal. He used a white tablecloth to provide contrast between dark colored LEGOs and the table.

4.2 Creative Ideation in Different Forms

Participants explored creative ideas by drawing on paper, using digital tools, and directly manipulating materials. Some sketched their ideas with pen and paper because their residual vision allowed them to and they were used to this technique. For a ceramic dragon mask, P18 first drew an outline of the dragon face to gauge where and what size to make the scales, moustache, eyes, and nose. She then showed the sketch to her instructor for feedback. P12 explored with beads and broken earrings while customizing eyeglass frames,

stating: *“I just kind of start laying stuff out before gluing it, . . . and see if I like it, seeing what needs to change. And then I’ll glue it down and then kind of just flow from there. . . . Just kind of like loosely put them on. . . . So if I don’t like the color, I can just grab some nail polish remover, and take it off.”* In contrast, others preferred testing ideas with digital tools. P5 said, *“I don’t love paper. Even when, when drawing things, I’ll gravitate towards a computer because then I can zoom in on them.”*

4.3 Feedback Solicitation and Iterative Improvements

Participants used initial ideation rounds to get opinions from others. When crocheting a football theme blanket, P11 showed her idea to her siblings to *“see if it looked centered and everything, if something was wrong with it, then double check before I turn it into yarn, to make sure it was legible, the words.”* P8 posted her custom Furby designs to online forums for Furby crafters to get critiques.

Participants in this study were open to receiving feedback on their ideas, especially from sighted people, for improvement. They broadly sought feedback from family members, friends, fellow crafters in the crafting community, and people with more experience in any particular aspect of the craft. Feedback was used to improve ideas, but also to check on visual aspects. P7 shared, *“I needed that sighted feedback to basically see if I was doing it correctly. And then once, once I got that feedback, then I can make my corrections.”* Similarly, P8 asked for color feedback after she created a base design to make sure colors matched as she desired, explaining that she was partially color blind. For example, one time she meant to create an orange and purple themed Furby, but instead the colors were a garish orange and brown until someone on a forum caught the mistake.

Surprisingly, participants had mixed feelings about receiving positive feedback. Some attributed positive feedback to people *“being too polite,” “wanting to be nice,”* and not wanting to be mean. They expressed skepticism and discomfort about positive feedback. P6 had sold over 300 embroidered Christmas stockings and key chains on Etsy, an ecommerce website, over the last one and a half years, yet she was surprised to receive all five-star reviews, stating, *“I honestly think my work isn’t really all that great.”* A lack of confidence in one’s creative abilities seems not uncommon, but participants were suspicious that people were merely sympathetic rather than impressed with their creative abilities. For instance, P15 clarified her skepticism, *“do you actually think it’s good? Or are you just afraid of saying that it’s not good? Or do you think it’s only good because I made it, and then if somebody else made it, it wouldn’t be good.”* P16 received comments like *“oh, it’s really cool or like, oh, I didn’t know you could draw that well or wow, this is really good.”* Although participants could not definitively say if these comments were overly positive due to their vision, they indicated that they felt such feedback was not always genuine. Overall, participants preferred critical feedback to help them improve, but they often had to specifically ask for it.

4.4 Creative Collaboration

Participants had experience collaborating with others in creative activities. Often, they were the only blind and low vision participant in these activities, and they engaged in collaboration differently from individual projects due to dynamics and interactions with sighted people. Most notably, when working in mixed-ability groups, participants divided tasks depending on capability: P15 had her partner finish up parts of a cosplay costume, such as the ears of a head piece, as she had trouble finding the center of objects. Creative collaboration was rooted in building ideas through communication and ability to perform assigned tasks, specifically, participants often worked around traditional division of labor due to accessibility issues (*i.e.*, sighted collaborators did tasks that required visual adjusting). In such cases, participants sought assistance from sighted partners to complete tasks that needed visual confirmation. In addition, participants experienced challenges in collaborative crafting because *“[sighted people] will be very much detail oriented in the sense that they want every little flick to be perfect”* (P2). To account for these differences, P2 focused on non-visual aspects such as texture because they *“have the same amount of context”* (P2) of the world. When participating in the LEGO city building project, P5 was afraid of his piece of the city *“standing out”* because he felt it was not *“quite as detailed or elegant”* as the rest of the pieces built by sighted group members.

5 DISCUSSION

We conducted an interview study to investigate how blind and low vision crafters selected and used materials to explore creative ideas. Although participants used many craft-specific materials, we found they also preferred low-cost and disposable materials with tactile and tentative (*“throw-away,”* modifiable) properties. Digital tools were not always the most cost effective or accessible options. From others, participants sought feedback for visual aspects, *i.e.*, alignment and colors, and they solicited critiques for improvement. But, they regarded feedback with caution: some participants questioned overly positive and sympathetic compliments as genuine. On the whole, despite prior work highlighting challenges with communication differences across mixed ability groups [4], participants had positive experiences with mixed-ability collaborations, often balancing sighted help for visual aspects of the work and contributing creatively in other areas of strength.

In line with our goal to learn from analog practices [6], we found that participants used crafting strategies comparable to sketching in HCI: they experimented with low-cost materials, explored ideas in quick iteration (and were as quick to throw out ideas), they incorporated feedback from others, and made multiple versions before committing to a final artifact [5]. While traditional sketching practices involve pen and paper [5], participants preferred materials with a tactile quality with definitive textures and that could be manipulated via haptics/touch.

Participants felt their ideas were embraced in collaboration with others, in contrast with prior work in which dominant group members made decisions instead of considering others’ opinions [4]. We note several key discrepancies: our participants were usually the only vision impaired group member, while co-design groups in prior work consisted of all blind and low vision members [4]. Also,

our participants were mainly experts in their craft, so they were likely more familiar with the creative process than participants in prior work without design experience [4]. Thus, we consider that context, mixed-ability group dynamics, and individual expertise may enhance blind and low vision collaborative experiences.

Based on our findings, we suggest that materials used for creative work with blind and low vision participants should: (1) meet specific purposes in the creative process; (2) be easy to obtain and be inexpensive; (3) have textures or shapes distinguishable by touch; (4) be tentative and temporary such that modifications are quick and easy. Further, our findings on creative analog tasks and endeavors contribute knowledge that may be useful in design work with blind and low vision co-designers and users, such as in design workshops and participatory design activities. Future work with blind and low vision creatives should use clearly organized systems and provide ample and high contrast workspaces.

6 LIMITATIONS

Our study is limited by crafting types explored and a focus on experts, with broad and limited technical applicability. Our questionnaire included responses from blind crafters, but we were unable to recruit blind creatives for interviews, so we cannot know how they approach crafts. We had limited ability to observe crafting and how participants worked. Due to recruitment challenges, we were unable to recruit more men participants, our findings are not necessarily representative of this participant imbalance. No craft activities that we explored resembled user interface design.

7 CONCLUSION

We report on an interview study with blind and low vision crafters to learn about their expertise in manipulating analog materials in creative endeavors. We found that participants used a wide variety of materials that prioritizes craft-specificity, low-cost, disposability, and tactility. Additionally, we found that participants employed techniques comparable to HCI design techniques, like sketching: by ideating in quick iteration, throwing out discarded ideas, and seeking feedback from others. Further, we found that collaboration with sighted colleagues could leverage mixed-ability benefits, but also be fraught with overly sympathetic feedback, rather than critiques focused on improving the work. We provide suggestions on the selection and use of materials for creative activities with blind and low vision creators, which can also benefit planning HCI research studies such as design workshops.

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APPENDIX I. PRELIMINARY QUESTIONNAIRE

Thank you for your interest in this study. We are inviting you to participate in this questionnaire about crafting and art projects or similar creative activities. This questionnaire will take about 3 - 4 minutes of your time. Your participation is anonymous and voluntary. Completing the questionnaire indicates your consent. If you do not consent to completing this questionnaire you may exit at any time by closing the questionnaire. We appreciate your time and willingness to support this study.

1. What's your occupation?
2. How would you best describe your vision?
- 3 Which of the following assistive devices if any that you use on a regular basis. Please select all that apply.

- Handheld magnifier
- Magnifying software
- Braille display
- Screen reader
- White cane and/or guide dog
- Other _____

4. What crafting or creative pursuits or activities do you enjoy?

5. What materials or tools do you use in the activities above?

Please indicate which of the following 20 materials you found to be effective in any creative activities you participated in. Please select all that apply.

- Cardboard
- Felt
- Construction paper
- Printer paper
- Index cards
- Foam boards
- Popsicle sticks
- Cork stoppers
- Clay
- Rubber bands
- Cotton balls
- LEGO blocks
- Toothpicks
- Play-Doh
- Pipe cleaners
- Yarn or thread
- Glue
- Scotch tape
- Nail polish
- Scissor
- Other (you may list as many items as you like)

APPENDIX II. INTERVIEW QUESTIONS

Questions include, but are not limited to, the following:

Crafting Activities

1. How are you today?
2. How would you best describe your vision?
3. What crafting activities do you enjoy? (*note which one requires crafting materials and go back later*)

4. What is your favorite crafting activity? Why?

5. How long have you been doing (activity in Q3)? How did you first learn how to (activity in Q3)?

6. How did you make sure you properly followed the instructions?

7. When was the last time you (activity in Q3), can you tell me about how you (activity in Q3) the last time?

8. What materials did you use the last time you (activity in Q3)?

9. Are these the materials you usually use for (activity in Q3)?

10. Why do you choose these materials (in 6.2)?

11. How do you organize these materials?

12. Did you complete your craft (in Q3) in one sitting or did you revisit it multiple times to finish it? Please tell me more about your process.

Techniques and "Work-arounds"

1. What do you think are difficult about (activity in Q3)? What did you do to improve your skills?

2. (*If you have difficulties in (activity in Q3), what did you usually do about them? Did you involve other people to try to solve the problems, for example, calling a friend for help?*)

Feedback

1. Did you share your crafts (created in Q3) with other people? How did you usually do that?

2. What feedback did you receive and did you make changes to your crafts based on the feedback? Why?

3. (*If given away*) Did people tell you something like they wish something would be different for (the object)? What did you think about that and how did you respond?

4. Did you make customized crafts because other people requested? What were they? How did you make sure you made the artifact the way they wanted?

5. Did you join any communities/hobby groups for (activity in Q3)?

- a. What did you usually do in these groups/communities?

- b. How did you interact with other members in these communities/groups? (In person or online?)

- c. Did you ask people in the same group for feedback on your crafts? How did you share your crafts? (send pictures? text?)

6. Did you participate in any crafting activities with other people? What kind are they?

- a. If yes, please tell me about a time you participated in crafting activities with other people? What did you do in the group? (Are there other blind people in the group?)

- b. Did you experience any challenges? Please describe.

- c. Did you receive feedback on what you created? What feedback did you get? What did you do? Did you make changes or improvements to your crafts based on the feedback.

- d. (*if mention received feedback they didn't like, or*) Did you receive feedback you did not like, for example, feedback that was irrelevant or overly critical? What did you do?

7. Have you worked in any kind of non-crafting focused projects/committees with other people?

- a. If yes, what kind of project/committee is it? Please tell me what happened.

- b. What did you do in the group?

c. Did you receive feedback on what you did for the project? What feedback did you get? What did you do? Did you make changes or improvements to your work based on the feedback.

d. (if mention received feedback they didn't like, or) Did you receive feedback you did not like, for example, feedback that was irrelevant or overly critical? What did you do?

8. I am curious about the dynamic/interactions between sighted and blind/low vision people such as the style of work in mixed groups. Anything comes to your mind?

APPENDIX III. MATERIALS USED BY BLIND AND LOW VISION PARTICIPANTS

In the questionnaire, respondents were asked about two types of materials (1) craft specific materials that they used specifically related to their craft activities, and (2) commonly used materials that might be low-cost, household items. See Appendix I for how these questions were worded. Our question about (1) craft specific materials was open ended since we could not know in advance

what crafts respondents did. Our question about (2) included a list of materials and an “other” option to add their own. Below are the responses we received for each question.

(1) Craft Specific Materials

aluminum, artificial birds, basic Furby toy, batting materials, bias tape, canvas, charcoal, chemical perfumes, clay, colorants, doorknobs, fabric, fiber, flannel, Furby eye chips, glaze, linen, lye, metal rod, mica, microchips, origami paper, paint thinner, all sorts of seed oil, sheet metal, steel, tracing paper, vegetable oils, wool

(2) Commonly Used Craft Materials (responses as marked, or added in “other” category)

3D stickers, acrylic paints, beads, branches, buttons, candle wax, crystals, cardboard, cloth, construction paper, craft sparkles, cupcake liners, feathers, felt, foam, food coloring, gems and parts from broken jewelry, glitters, glue, grass, hot glue, leaves, LEGO, magazines, Mod Podge, nail supplies (e.g., nail polish, nail glue, and fake nails), paper, pennies, Play-Doh, recycled materials, rubber, sand, spray paint, spray adhesive, sticker dots, sticks, tin foil, toilet paper rolls, toothpicks, twigs, yarn, wine corks, wood, zip tie